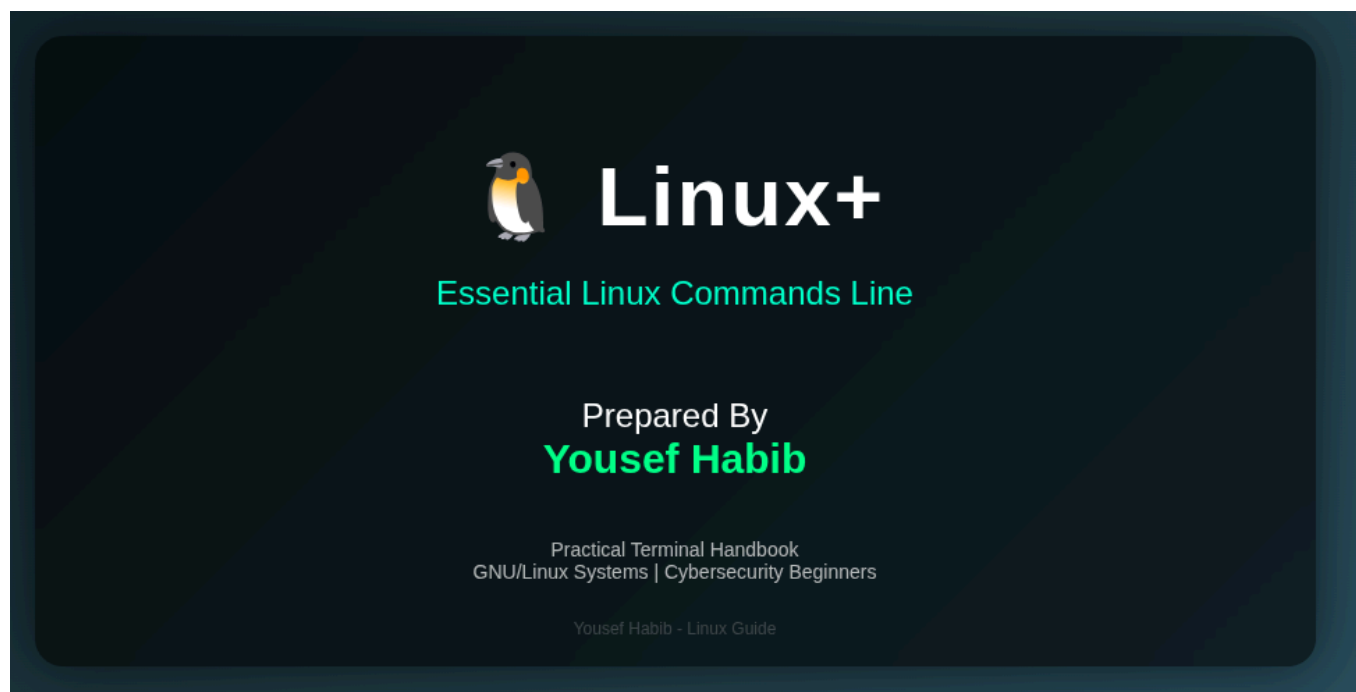


Linux Fundamentals



Linux Fundamentals

sudo su ⇒ root كاملة التحويل من المستخدم الحالي إلى مستخدم

```
—(yousef_7abib@kali) - [~/Desktop]
└─$ sudo su
[sudo] password for kali:
└─(root@kali) - [/home/kali/Desktop]
└─#
```

sudo apt install ⇒ الكالي من اصدار احدث انزل علشان

whoami ⇒ اي اسمك ولا يوزر ولا ادمن النظام ف مين انت تعرف علشان

```
└─(yousef_7abib@kali) - [~/Desktop]
└─$ whoami
yousef_7abib
```

ls ⇒ يعرض الملفات والمجلدات الموجودة في المسار الحالي

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ls
bandit.txt hashes 'WhatsApp Image 2024-08-30 at 16.57.
26_2651334d.jpg'
```

pwd ⇒ يعرض المسار الحالي الذي تعمل بداخله في النظام

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ pwd
/home/kali/Desktop
```

ls -lah ⇒ كل صلاحيات اعراف كمان وعلشان النظام في المخفيه الملفات اعراف علشان

فايل

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ls -lah
total 12K
drwxr-xr-x. 1 yousef_7abib yousef_7abib 124 Mar  1 18:57 .
drwx----- 1 yousef_7abib yousef_7abib 652 Mar  3 03:04 ..
-rw-r--r--. 1 yousef_7abib yousef_7abib 460 Feb 22 17:03 bandit
drwxr-xr-x. 1 yousef_7abib yousef_7abib 148 Mar  1 18:51 CommandFiles
-rw-r--r--. 1 yousef_7abib yousef_7abib   0 Feb 26 02:23 hashes
-rw-r--r--. 1 yousef_7abib yousef_7abib  53 Feb 19 21:34 CommonedTool.txt
-rw-r--r--. 1 yousef_7abib yousef_7abib 748 Feb 22 16:13 Ignited.txt
```

cat ⇒ الملفات اقرأ علشان

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat habib
```

```
my name is habib
```

touch ⇒ ملفات اعمل علشان

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ touch habib
```

mkdir ⇒ directories او فولدرات اعمل علشان

```
(yousef_7abib@kali) - [~/Desktop]
└─$ mkdir yousefhabib
```

akhs kntrol w as kntrol b ahfth w bcdha foldr jwa akth elshan => nano

```
(yousef_7abib@kali) - [~/Desktop]
└─$ nano habib
```

rm => mlf amsh elshan

apropos scan => scan btعمل اداه مثلا اعمل عاوز ولو

```
(yousef_7abib@kali) - [~]
└─$ apropos scan
airscan-discover (1) - Discover sane-airscan compatible scanners
alphasort (3) - scan a directory for matching entries
arp-scan (1) - Send ARP requests to target hosts and display response
awk (1) - pattern scanning and processing language
bulk_extractor (1) - Scans a disk image for regular expressions and other
dirb (1) - Web Content Scanner
dnsrecon (1) - DNS Enumeration and Scanning Tool
dpkg-scanpackages (1) - create Packages index files
```

```
dpkg-scansources (1) - create Sources index files
fc-scan (1) - scan font files or directories
```

```
(yousef_7abib@kali) - [~/Desktop]
└─$ rm habib
```

rm -r => mlf amsh elshan directory

```
(yousef_7abib@kali) - [~/Desktop]
└─$ rm -r yousefhabib
```

cp => mlf ansx elshan

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ cp bandit.txt home
```

mv ⇒ لمان مكان من ملف انقل علشان

echo ⇒ البرمجه في كده print امرزي

```
—(yousef_7abib@kali)-[~]
└─$ echo "Hi I'm Yousef Habib"
Hi I'm Yousef Habib
```

وتبدله كله الجوا بتشيل > اما يتمسح الجوا الكلام ما بدون الملف جوا نص تحط
<< ⇒ علشان
بالعايزه

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ echo "Hi I'm Yousef Habib" >> home
```

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ cat home
Hi I'm Yousef Habib
```

To save it in home file

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ echo "Hi I'm Yousef Habib" > home
```

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ cat home
Hi I'm Yousef Habib
```

ifconfig ⇒ الخاصة بالجهاز IP يعرض معلومات كروت الشبكة وعناوين

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ ifconfig
```

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.45.134 netmask 255.255.255.0 broadcast
192.168.45.
inet6 fe80::b21:627a:5bdc:e3ec prefixlen 64 scope
```

```
eid 0x20<link>
ether 00:0c:29:00:cf:ad txqueuelen 1000 (Ethernet)
RX packets 3923 bytes 663244 (647.6 KiB)
```

```
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 2766 bytes 238140 (232.5 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 4 bytes 240 (240.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 4 bytes 240 (240.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

اي قد مستهلك وانت الرامات تفاصيل بتعرضلك ⇒ free

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ free
total used free shared
buff/cache available
Mem: 2012816 705724 722908 8608
767696 1307092
Swap: 1048572 0 1048572
```

بتاعك هارد عن تفاصيل يجبلك علشان ⇒ df -h

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ df -h
Filesystem Size Used Avail Use% Mounted on
udev 942M 0 942M 0% /dev
tmpfs 197M 1.3M 196M 1% /run
/dev/sda1 79G 15G 60G 21% /
tmpfs 983M 0 983M 0% /dev/shm
```

```
tmpfs 5.0M 0 5.0M 0% /run/lock
```

```
tmpfs 197M 124K 197M 1% /run/user/
```

بروسیسیور عن اکثر معلومات اعرفك علشان $\Rightarrow$ ps aux

```
(yousef_7abib@kali)-[~/Desktop]
└─$ ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START
T TIME COMMAND
root 1 0.0 0.6 21592 12648? Ss 04:
2 0:04 /sbin/init
root 2 0.0 0.0 0 0? S 04:
2 0:00 [kthreadd
root 3 0.0 0.0 0 0? S 04:
2 0:00 [pool_wor
root 4 0.0 0.0 0 0? I< 04:
2 0:00 [kworker/
root 5 0.0 0.0 0 0? I< 04:
2 0:00 [kworker/
root 6 0.0 0.0 0 0? I< 04:
2 0:00 [kworker/
root 7 0.0 0.0 0 0? I< 04:
2 0:00 [kworker/
root 9 0.0 0.0 0 0? I< 04:
2 0:00 [kworker/
root 11 0.0 0.0 0 0? I 04:
2 0:00 [kworker/
root 12 0.0 0.0 0 0? I< 04:
2 0:00 [kworker/
root 13 0.0 0.0 0 0? I 04:
2 0:00 [rcu_task
root 14 0.0 0.0 0 0? I 04:
2 0:00 [rcu_task
root 15 0.0 0.0 0 0? I 04:
2 0:00 [rcu_task
root 16 0.0 0.0 0 0? S 04:
```

```
2 0:00 [ksoftirq
root 17 0.1 0.0 0 0? I 04:
2 0:06 [rcu_pree
root 18 0.0 0.0 0 0? S 04:
2 0:00 [migratio
root 19 0.0 0.0 0 0? S 04:
2 0:00 [idle_inj
```

```
root 20 0.0 0.0 0 0? S 04:
2 0:00 [cpuhp/0]
```

chmod ⇒ اغير علشان permissions

r = 4 ⇒ الملف اقرا علشان

w = 2 ⇒ الملف جوا اكتب اعراف علشان

x = 1 ⇒ اعمل اعراف علشان execution

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ls -lah
total 12K
drwxr-xr-x. 1 yousef_7abib yousef_7abib 124 Mar  1 18:57 .
drwx----- 1 yousef_7abib yousef_7abib 652 Mar  3 03:04 ..
-rw-r--r--. 1 yousef_7abib yousef_7abib 460 Feb 22 17:03 bandit
drwxr-xr-x. 1 yousef_7abib yousef_7abib 148 Mar  1 18:51 CommandFiles
-rw-r--r--. 1 yousef_7abib yousef_7abib   0 Feb 26 02:23 hashes
-rw-r--r--. 1 yousef_7abib yousef_7abib  18 Mar  3 03:38 home
-rw-r--r--. 1 yousef_7abib yousef_7abib  53 Feb 19 21:34 CommonedTool.txt
-rw-r--r--. 1 yousef_7abib yousef_7abib 748 Feb 22 16:13 Ignited.txt
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ chmod 744 home
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ls -lah
total 12K
drwxr-xr-x. 1 yousef_7abib yousef_7abib 124 Mar  1 18:57 .
drwx----- 1 yousef_7abib yousef_7abib 652 Mar  3 03:04 ..
-rw-r--r--. 1 yousef_7abib yousef_7abib 460 Feb 22 17:03 bandit
drwxr-xr-x. 1 yousef_7abib yousef_7abib 148 Mar  1 18:51 CommandFiles
-rw-r--r--. 1 yousef_7abib yousef_7abib   0 Feb 26 02:23 hashes
-rwxr--r--. 1 yousef_7abib yousef_7abib  18 Mar  3 03:38 home
-rw-r--r--. 1 yousef_7abib yousef_7abib  53 Feb 19 21:34 CommonedTool.txt
-rw-r--r--. 1 yousef_7abib yousef_7abib 748 Feb 22 16:13 Ignited.txt
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ chmod +x home
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ls -lah
```

```
total 12K
drwxr-xr-x. 1 yousef_7abib yousef_7abib 124 Mar  1 18:57 .
drwx----- 1 yousef_7abib yousef_7abib 652 Mar  3 03:04 ..
-rw-r--r--. 1 yousef_7abib yousef_7abib 460 Feb 22 17:03 bandit
drwxr-xr-x. 1 yousef_7abib yousef_7abib 148 Mar  1 18:51 CommandFiles
-rw-r--r--. 1 yousef_7abib yousef_7abib   0 Feb 26 02:23 hashes
-rwxr-xr-x. 1 yousef_7abib yousef_7abib  18 Mar  3 03:38 home
-rw-r--r--. 1 yousef_7abib yousef_7abib  53 Feb 19 21:34 CommonedTool.txt
-rw-r--r--. 1 yousef_7abib yousef_7abib 748 Feb 22 16:13 Ignited.txt
```

execution for all اضيف #علشان

الباشون ب المكتوبه الملفات تقرألي علشان ⇒ python 3

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ python3 ./python
Hi i'm Yousef Habib
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ./python
zsh: permission denied: ./python
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ sudo ./python
sudo: ./python: command not found
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ls -lah
total 12K
drwxr-xr-x. 1 yousef_7abib yousef_7abib 124 Mar  1 18:57 .
drwx----- 1 yousef_7abib yousef_7abib 652 Mar  3 03:04 ..
-rw-r--r--. 1 yousef_7abib yousef_7abib 460 Feb 22 17:03 bandit
drwxr-xr-x. 1 yousef_7abib yousef_7abib 148 Mar  1 18:51 CommandFiles
-rw-r--r--. 1 yousef_7abib yousef_7abib   0 Feb 26 02:23 hashes
-rwxr-xr-x. 1 yousef_7abib yousef_7abib  18 Mar  3 03:38 home
-rw-rw-r-- 1 yousef_7abib yousef_7abib 18 Mar 27 06:36 python
-rw-r--r--. 1 yousef_7abib yousef_7abib  53 Feb 19 21:34 CommonedTool.txt
-rw-r--r--. 1 yousef_7abib yousef_7abib 748 Feb 22 16:13 Ignited.txt
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ chmod +x python
```



```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ./python
Hi i'm Yousef Habib
nano python =>
#!/usr/bin/env python3
print("Hi i'm Yousef Habib")
```

كبير ملف من كلمه اقتصف علشان ⇒ grep

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat home | grep "Habib"
Hi I'm Yousef Habib
```

ده الملف سطور عدد تعدلي علشان ⇒ wc

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat home | wc
1 4 23
```

to write by using bash script ⇒ nano file then #!/bin/bash ⇒ write your comands ⇒ ctrl s+ctrl x bash file.sh

```
#!/bin/bash
pwd
echo "I'm Yousef Habib"
└─(yousef_7abib@kali)-[~/Desktop]
└─$ touch habin
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ chmod +x habib
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ./habib
/home/yousef_7abib/Desktop
I'm Yousef Habib
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat habib
#!/bin/bash
```

```
pwd
echo "I'm Yousef Habib"
```

```
#!/bin/bash
pwd
echo "I'm Yousef Habib"
file="/home/yousef_7abib/Desktop/home"
cat "$file"
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ ./habib
/home/yousef_7abib/Desktop
I'm Yousef Habib
Hi I'm Yousef Habib
```

to make a for loop in linux use $\Rightarrow$ for i in 0 .. 100 ; do echo "\$i" ; done

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ for i in {1..5} ; do echo "$i" ;done
1
2
3
4
5
```

to print each line using for loop $\Rightarrow$

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ for i in `cat home` ; do echo "$i" ;done
Hi
I'm
Yousef
Habib
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ for i in `cat home` ; do echo "$i habib" ;done
Hi habib
I'm habib
habib habib
Yousef Habib
```

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ for i in {1..25}; do touch file{1..24};done
```

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ ls
habib file17 file
ahmed.sh file18 file
bandit.txt file19 file
file1 file2 file
file10 file20 file
```

file11 file21 hashes
file12 file22 home
file13 file23 python
file14 file24 'WhatsApp Image 2024-08-30 at 16.57.
26_2651334d.jpg'
file15 file
file16 file

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ rm file*
```

استخدام ب الارقام يحددلي علشان grep

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ cat hashes |grep -E "[0-9]"
202cb962ac59075b964b07152d234b
0e7517141fb53f21ee439b355b5a1d0a
d9b1d7db4cd6e70935368a1efb10e
7363a0d0604902af7b70b271a0b
a665a45920422f9d417e4867efdc4fb8a04a1f3fff1fa07e998e86f7f
a27ae
```

⇒ الحروف يحددلي علشان

```
—(yousef_7abib@kali)-[~/Desktop]
└─$ cat hashes |grep -E "w"
202cb962ac59075b964b07152d234b
0e7517141fb53f21ee439b355b5a1d0a
d9b1d7db4cd6e70935368a1efb10e
7363a0d0604902af7b70b271a0b
```

a665a45920422f9d417e4867efdc4fb8a04a1f3fff1fa07e998e86f7f
a27ae

⇒ كله الكلام يحددلي علشان

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat hashes |grep -E "."
202cb962ac59075b964b07152d234b
0e7517141fb53f21ee439b355b5a1d0a
d9b1d7db4cd6e70935368a1efb10e
7363a0d0604902af7b70b271a0b
a665a45920422f9d417e4867efdc4fb8a04a1f3fff1fa07e998e86f7f
a27ae
```

علامه اي او (ال او @ال علامه يحددلي علشان

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat bandit.txt |grep -E "\\@"
```

```
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat bandit.txt |grep -E "\\("
└─(yousef_7abib@kali)-[~/Desktop]
└─$ cat hashes |grep -E "\\s" => علشان يحدد المسافات
```

الي الادوات كل يجيلي 2 ولا حرف اول مثلا وكتبت اداه ناسي لو علشان ⇒ apropos

بيها مرتبطه

```
└─(yousef_7abib@kali)-[~]
└─$ apropos who
LS (6) - display animations aimed to correct
users who acci...
ldapwhoami (1) - LDAP who am i? tool
Net::Whois::IP (3pm) - Perl extension for looking up the w
hois informatio...
rwho (1) - who is logged in on local machines
rwhod (8) - system status server
sl (6) - display animations aimed to correct
users who acci...
w (1) - Show who is logged on and what they
are doing.
```

who (1) - show who is logged on
whoami (1) - print effective user name
whois (1) - client for the whois directory service
whois.conf (5) - alternative WHOIS servers list for whois client

what is ⇒ بتجيبك عن تفاصيل شويه command

```
—(yousef_7abib@kali)-[~]  
└─$ whatis nmap  
nmap (1) - Network exploration tool and security / port scanner
```

info ⇒ برده الاداه عن تفاصيل تجيبك علشان

man ⇒ برده الاداه عن تفاصيل تجيب

info -w ⇒ للكوماند execute البيعمل للملف توصل علشان

```
—(yousef_7abib@kali)-[~]  
└─$ info -w ssh  
*manpages*
```

```
—(yousef_7abib@kali)-[~]  
└─$ info -w nmap  
*manpages*
```

```
—(yousef_7abib@kali)-[~]  
└─$ info -w ls  
/usr/share/info/coreutils.info.gz
```

dd ⇒ copying from path to path

```
—(yousef_7abib@kali)-[~/Desktop]  
└─$ dd if=/home/kali/Desktop/hashes of=/tmp/hashes_copy  
0+1 records in
```

0+1 records out

197 bytes copied, 0.000224868 s, 876 kB/s

What we can do Syntax Real example of syntax

Find files based on
filename

```
find [directory path] -type f -name "filename"
```

```
find /home/Andy -type f -name  
sales.txt
```

Find Directory
based on directory
name

```
find [directory path] -type d -name [filename]
```

```
find /home/Andy - type d - name  
pictures
```

Find files based on
size

```
find [directory path] -type f -size [size]
```

```
find /home/Andy -type f -size 10 c  
(c for bytes,  
k for kilobytes  
M megabytes  
G for gigabytes  
type:'man find' for full information on  
the options)
```

Find files based on
username

```
find [directory path] -type  
f - user [username]
```

```
find /etc/server -type f -user habib
```

Find files based on
group name

```
find [directory path] -type  
f - group [group name]
```

```
find /etc/server -type f -group  
teamstar
```

Find files modified
after a specific
date

```
find [directory path] -type f -newermt '[date and  
time]'
```

```
find / - type f - newermt ' 6 / 30 / 2020  
0 00 00 '  
(all dates/times after 6 / 30 / 2020  
0 00 00 will be considered a  
condition to look for)
```

Find files based on
date modified

```
find [directory path] - type  
f - newermt [start date  
range]! - newermt [end  
date range]
```

```
find / - type f - newermt 2013 09 12!
```

- newermt 2013 09 14
(all dates before 2013 09 12 will be
excluded; all dates after 2013 09 14
will be excluded, therefore this only

leaves 2013 09 13 as the date to look for.)

Find files based on date accessed

```
find [directory path] - type  
f - newerat [start date
```

```
find / - type f - newerat 2017 09 12! -  
newerat 2017 09 14
```

```
range]! - newerat [end  
date range]
```

(all dates before 2017 09 12 will be excluded; all dates after 2017 09 14 will be excluded, therefore this only leaves 2017 09 13 as the date to look for.)

Find files with a specific keyword

```
grep -iRl [directory/path/keyword] grep^ - iRl^ '/folderA/flag'
```

read the manual for the find command man^ find man^ find

الاجري الاوامر بعض

What we can do Syntax Real example of syntax

copy file/folder

```
cp [filename/folder]  
[directory]  
(remember, if the  
filename/folder name has  
spaces then you will need to
```



```
encase the filename with  
speech marks such as cp "  
[filename with spaces]"  
[directory]. This applis to  
other commands such as mv.  
)
```

```
cp ssh.conf /home/newfolder
```

move file/folder mv [filename] [directory] mv ssh.conf /home/newfolder

move multiple
files/folders
simultaneously

```
mv [file 1 ] [file 2 ] [file 3 ] - t  
[directory to move to]
```

```
mv logs.txt keys.conf script.py - t  
/home/savedWork
```

Move all files from
current directory
into another
directory

```
mv * [directory to move files  
to] mv^ *^ /home/scripts
```

rename
files/folder

```
mv [current filename] [new  
filename] mv^ ssh.conf^ NewSSH.conf
```

create a file touch [filename] touch newFile.txt

create a folder mkdir [foldername] mkdir newFolder

```
open file for  
editing nano^ [filename] nano^ keys.conf
```

```
output contents of  
file cat^ [filename] cat^ keys.conf
```

upload file to a
remote machine

```
scp [filename]  
[usernameIP of remote  
machine ]:/[directory to  
upload to]
```

```
scp example.txt  
habib@ 192. 168. 100. 123 /home/habib/
```

```
run an bash script  
program ./[name^ of^ script] ./timer  
open a file for  
reading/editing nano^ [filename] nano^ readME.txt
```

Regex ⇒

short cuts for kali ⇒

what is PATH? ⇒ اللينكس ف الي مسارات بيجهلك دا

We can modify our **PATH** and **LD_LIBRARY_PATH** (shared libraries) environment variables to use these trusted binaries:

Modifying Environment Variables to Include Trusted Paths

```
investigator@MACHINE_IP:~$ export PATH=/mnt/usb/bin:/mnt/u  
sb/sbininvestigator@MACHINE_IP:~$ export LD_LIBRARY_PATH=/  
mnt/usb/lib:/mnt/usb/lib64
```

Users And Groups

to change between users ⇒ su - [username]

sudo useradd [username]

To create a user with a home directory and specific shell:

```
sudo useradd -m -s /bin/bash [username]
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo useradd -m -s /bin/bash hr
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo cat /etc/shadow  
hr:!:20044:0:99999:7:::  
└─(yousef_7abib@kali)-[~]  
└─$ sudo passwd hr  
New password:  
Retype new password:  
passwd: password updated successfully
```

```
└─(yousef_7abib@kali)-[~]  
└─$ su - hr  
Password:  
└─(yousef_7abib@kali)-[~]  
└─$ whoami  
yousef_7abib
```

Deleting Users: To delete a user account: ⇒ محذفتش بس اليوزر هنا حذف انت

بتاعته الفولدرات

```
sudo userdel [username]
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo useradd hr  
[sudo] password for kali:
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo userdel hr
```

To remove a user and their home directory: ⇒ الفولدرات و اليوزر هنا حذف انت

بتاعته

```
sudo userdel -r [username]
```

Modifying Users: Use usermod to modify existing user settings ⇒ صلاحيات تعلي

اليوزر

```
sudo usermod [options] [username]
```

Locking user : to make lock for user ⇒ علي حابه يعمل انه من يوزر تحظر علشان

usermod -L [username]

Creating Groups: To create a new group: ⇒ جروب تضيف علشان

sudo groupadd [groupname]

```
└─(yousef_7abib@kali)-[~]
└─$ sudo groupadd ayhaga
```

```
└─(yousef_7abib@kali)-[~]
└─$ sudo cat /etc/group
ayhaga:x:1002:
```

Deleting Groups: To delete a group: ⇒ جروب تمسح علشان

sudo groupdel [groupname]

Modifying Groups: To add a user to a group, use usermod with the - aG option:

⇒ جروب ل يوزر تضيف علشان

sudo usermod - aG [groupname] [username]

```
└─(yousef_7abib@kali)-[~]
└─$ sudo usermod -aG kali hr
```

```
└─(yousef_7abib@kali)-[~]
└─$ groups hr
hr : hr kali
```

⇒ for checking that user has been added to each groups

groups [username]

```
└─(yousef_7abib@kali)-[~]
└─$ groups hr
hr : hr kali
```

Display All Users: You can view all users by examining the /etc/passwd file:

cat /etc/passwd

Display All Groups: View all groups by examining the /etc/group file:

cat /etc/group

⇒ جديد جروب ل قديم جروب من الجروب اسم اغير علشان:

`sudo groupmod -n new_group_name old_group_name`

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo groupmod -n ayhaga habib  
groupmod: group 'habib' does not exist
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo useradd habib
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo groupmod -n ayhaga habib  
groupmod: group 'ayhaga' already exists
```

```
└─(yousef_7abib@kali)-[~]
```

```
└─$ sudo groupdel ayhaga
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo groupmod -n ayhaga habib
```

```
└─(yousef_7abib@kali)-[~]  
└─$ sudo cat /etc/group  
ayhaga:x:1003:
```

Changing User and Group Ownership ⇒ لفايل الملكيه تغير علشان مثلا فايل ل روت ل يوزر مثلا من
`sudo chown [owner]:[group] [file/directory]`

```
└─(yousef_7abib@kali)-[~/Desktop]  
└─$ sudo chown hr:hr alex
```

```
└─(yousef_7abib@kali)-[~/Desktop]  
└─$ ls -lah  
total 96K  
drwxr-xr-x 2 yousef_7abib yousef_7abib 4.0K Nov 17 12:17.
```

```
drwx----- 20 yousef_7abib yousef_7abib 4.0K Nov 17 12:14 ..
-rwxrwxr-x 1 yousef_7abib yousef_7abib 88 Nov 2 02:21 habib
-rw-rw-r-- 1 yousef_7abib yousef_7abib 289 Nov 8 18:04 yousefhabib.
txt
-rwxrwxr-x 1 yousef_7abib yousef_7abib 67 Nov 2 02:17 ahmed.sh
-rwxrwxr-x 1 yousef_7abib yousef_7abib 21 Nov 12 06:15 ayhaga
-rw-rw-r-- 1 yousef_7abib yousef_7abib 412 Oct 18 07:29 bandit.txt
-rw-rw-r-- 1 yousef_7abib yousef_7abib 197 Oct 11 05:23 hashes
-rwxr-xr-x 1 yousef_7abib yousef_7abib 23 Oct 27 05:59 home
-rwxrwxr-x 1 yousef_7abib yousef_7abib 51 Oct 27 06:40 python
-rw-rw-r-- 1 hr hr 21 Nov 17 12:17 tefa
-rwxrw-rw- 1 yousef_7abib yousef_7abib 49K Aug 30 09:57 'WhatsApp Image
2025-08-30 at 16.57.26_2651334d.jpg'
```

/etc/skel ⇒ النظام بتاعه وال فولدرات الفايلس تشوف علشان

/etc/shadow ⇒ الباسورد يعرض علشان

/etc/passwd ⇒ اللينكس لنظام المستخدمين عن معلومات يعرضلي علشان

```
/etc/passwd
username:x:UID:GID:Full Name:/home/username:/bin/bash
/etc/shadow
username:$6$abcdefgh$xyz...:17801:0:99999:7:::
/etc/skel/
├─ .bashrc
├─ .bash_profile
├─ .profile
└─ .vimrc
```

المواقع عليه ارفع بيا خاص سيرفر اعمل علشان دا

بتاعي الموقع زار الي مين انا واعرف اشوف علشان دا

Ngrok ⇒ توصله الناس كل اخلي بتاعي الموقع احول علشان

منه توصلي تعرف الناس كل الي العام بتاعي بي الاي اجيب علشان

افتح localhost برورر علي

```
var/www/html/ المسار علي هخش
الصفحه علي اعدل علشان
```

لسه برايفت هيبقي بردو اعدل ما بعد

```
var/log/apache 2 / هخش  
علي بر دو  
واعمل cat access.log
```

بتاعي موقع علي دخول يسجل حاول مين هنا هيظهرلي

بعدها هتنزل ngrok بتاعك موقع تخلي علشان public

هتعمل وبعدها export بتاع للباث snap/bin

يوصله عارف مش هو علشان
فوق فالصوره موضح هو ما زي جدا عادي التول تشغل وبعدها

بتاعي ديستوب من ازوره و اللينك اخذ جربت لما windows

استخدام ب لنفسي سيرفر اعمل لله الحمد قدرت انا وبكده ngrok

How To Install a tools? ⇒

#apt install

#snap install

#dpkg -i <app.deb>

لا تنسوني من صالح دعائكم